

1. Solve for x:

$$4x + 15 = 3x + 33$$

- a. 16 b. 17 c. 18 d. 19 e. 20

2. One-half of a number increased by 12 is 36.
What is the number?

- a. 42 b. 46 c. 48 d. 50 e. 52

3. A father is 6 times as old as his son.
In 12 years, he will be 3 times as old.
How old is the son now?

- a. 3 b. 4 c. 5 d. 6 e. 7

4. Solve for x:

$$5(x - 4) = 3x + 16$$

- a. 12 b. 13 c. 14 d. 15 e. 18

5. A shirt was sold for ₱2,040 after a 15% discount.
What was its original price?

- a. ₱2,200 b. ₱2,300 c. ₱2,400 d. ₱2,500 e. ₱2,600

6. If $\frac{3}{4}$ of a number is 24 more than $\frac{1}{4}$ of the same number,
what is the number?

- a. 44 b. 46 c. 48 d. 50 e. 52

7. Solve for x:

$$3x + 8 = 2(x + 20)$$

- a. 28 b. 30 c. 32 d. 34 e. 36

8. A worker earns ₱950 per day. After working 24 days, he spent ₱8,800. How much money does he have left?

- a. ₱13,600 b. ₱14,000 c. ₱14,400 d. ₱14,800 e. ₱15,200

9. The ratio of boys to girls is 5:7.
If there are 144 students, how many girls are there?

- a. 74 b. 80 c. 84 d. 88 e. 90

10. Solve for x:

$$7x - 12 = 5x + 20$$

- a. 14 b. 15 c. 16 d. 17 e. 18

11. If 40% of a number is 84,
what is 60% of the same number?

- a. 116 b. 120 c. 126 d. 132 e. 138

12. The average of five numbers is 36.
Four of the numbers are 28, 34, 37, and 41.
What is the fifth number?

- a. 36 b. 38 c. 40 d. 42 e. 44

13. Solve for x:

$$4(x + 2) = 2(x + 14)$$

- a. 9 b. 10 c. 11 d. 12 e. 13

14. A merchant bought an item for ₱3,200.
He marked it up by 25% and then gave a 10% discount.
What was the selling price?

a. ₱3,500 b. ₱3,600 c. ₱3,700 d. ₱3,800 e. ₱3,900

15. A train travels 90 km/h.

How far will it travel in 2 hours and 40 minutes?

a. 220 km b. 230 km c. 240 km d. 250 km e. 260 km

16. Solve for x:

$$2x + 15 = x + 34$$

a. 17 b. 18 c. 19 d. 20 e. 21

17. If $\frac{5}{8}$ of a number is 20 less than the number itself,
what is the number?

a. 48 b. 52 c. 56 d. 60 e. 64

18. A project can be finished by 12 workers in 15 days.

How many days will it take 20 workers to finish the same project?

a. 7 b. 8 c. 9 d. 10 e. 11

19. Solve for x:

$$6(x - 2) = 4(x + 7)$$

a. 18 b. 19 c. 20 d. 21 e. 22

20. A salary of ₱28,000 was increased by 15%.

What is the new salary?

a. ₱31,200 b. ₱32,000 c. ₱32,200 d. ₱33,000 e. ₱33,200

21. A father is 5 times as old as his son.

In 15 years, he will be twice as old.

How old is the father now?

- a. 20 b. 25 c. 30 d. 35 e. 40

22. Solve for x:

$$8x + 12 = 5x + 51$$

- a. 11 b. 12 c. 13 d. 14 e. 15

23. If 35% of a number is 70,
what is 55% of the same number?

- a. 100 b. 105 c. 110 d. 115 e. 120

24. A bus traveled 420 km in 6 hours.
If it maintained the same speed,
how far would it travel in 8 hours?

- a. 520 km b. 540 km c. 560 km d. 580 km e. 600 km

25. Solve for x:

$$3(x + 8) = 2(x + 20)$$

- a. 12 b. 14 c. 16 d. 18 e. 20

26. The ratio of A:B:C is 2:3:5.
If their total is 300,
what is the value of C?

- a. 120 b. 140 c. 150 d. 160 e. 180

27. If $\frac{2}{3}$ of a number is 16 more than $\frac{1}{3}$ of the same number,
what is the number?

- a. 42 b. 45 c. 48 d. 51 e. 54

28. Solve for x:

$$9x - 18 = 6x + 30$$

- a. 14 b. 15 c. 16 d. 17 e. 18

29. A tank contains 1,800 liters.

If 35% is used and then 150 liters are added,
how much water is in the tank?

- a. 1,250 b. 1,300 c. 1,320 d. 1,350 e. 1,380

30. The average of six numbers is 42.

Five of the numbers are 35, 38, 41, 44, and 46.
What is the sixth number?

- a. 44 b. 46 c. 48 d. 50 e. 52

31. Solve for x:

$$5(x - 1) = 3(x + 11)$$

- a. 17 b. 18 c. 19 d. 20 e. 21

32. A train 180 meters long passes a pole in 12 seconds.

What is its speed in km/h?

- a. 48 b. 54 c. 60 d. 66 e. 72

33. If 18 is 15% of a number,
what is 40% of that number?

- a. 42 b. 44 c. 46 d. 48 e. 50

34. Solve for x:

$$4x + 25 = 3x + 41$$

a. 14 b. 15 c. 16 d. 17 e. 18

35. A merchant bought 80 notebooks for ₱25 each and sold them for ₱35 each.
What was the total profit?

a. ₱700 b. ₱750 c. ₱800 d. ₱850 e. ₱900

36. If 3 workers can finish a job in 24 days, how many days will 8 workers need?

a. 8 b. 9 c. 10 d. 11 e. 12

37. Solve for x:

$$7(x - 3) = 4x + 24$$

a. 10 b. 11 c. 12 d. 13 e. 14

38. A class has 200 students.
If 45% are boys,
how many girls are there?

a. 100 b. 105 c. 110 d. 115 e. 120

39. The sum of two numbers is 56.
One number is 8 more than the other.
What is the larger number?

a. 30 b. 31 c. 32 d. 33 e. 34

40. Solve for x:

$$\left(\frac{3}{4}\right)x + 6 = 30$$

a. 28 b. 30 c. 32 d. 34 e. 36